DOUGLAS COLLEGE
COMMERCE AND BUSINESS ADMINISTRATION

Reconstructing a Simulated Transaction Across Four Android Apps:

An Exploratory Comparison of Rooted and Non-Rooted Forensic Workflows

Project Proposal

CSIS 4440 – 003: Mobile CyberSecurity

Fall 2026

Submitted by:

Group 6

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GitHub Repository:

https://github.com/RafaellGramata/F26_4440_S003_G6

Submitted to:

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1. Project Overview and Motivation

In this project, we will investigate forensic artifacts from four Android applications to attempt to reconstruct simulated activities involving a suspicious transaction. In a controlled and fictional simulation, a suspect drafts a plan in Joplin, researches a meeting location using Firefox, communicates with a potential buyer through Telegram, and travels to the meeting point while recording the route with OpenTracks. We will be documenting the sequence of actions and compare it with events recorded by forensic examination. The goal of this project is the evaluation of the how the artifacts can be admissible to be considered as evidence.

Both members of the group will be using their own Android devices via emulators. Rafaell will obtain application data from a root-enabled device using ADB and analyze it with ALEAPP. Hoang will be using a non-rooted device using Avilla Forensics, including APK downgrade/backup acquisition functionality. The findings from both emulators having different android environments will combine into one fictional investigation on a shared timeline of events, with each artifact clearly identified by its source emulator and the forensic method of its acquisition.

2. Tools, Apps, and Novelties

Rafaell's Selection of Tool, Apps, and Novelty

Tool: Android Logs, Events, and Protobuf Parser (ALEAPP). This tool is free to use that contains an open-source parser that processes supported Android extraction artifacts. It is also able to generate forensic reports. The acquisition process will be done using ADB on a root-enabled Android emulator.

Apps:

- Joplin: A free open-source note-taking application to create, organize, and manage notes and notebooks.
- OpenTracks: A free open-source GPS tracking application to record outdoor activities, routes, geo locations, and movement data.

Novelty:

1. **Forensic Artifact Persistence Experiment**: For example, in using Joplin, the following are snapshotted: the creation of a note with sample data, the change of data of a note, and the deletion of a note. The logged records of all three snapshots are documented and compared to see which artifacts persist, change, or disappear after specific user actions. ALEAPP parses evidence, but it does not conduct this controlled experiment.
2. **Evaluation of ALEAPP's accuracy and completeness in forensics**: For example, in creating ten events using both Joplin and OpenTracks, the number of events identified by ALEAPP should also include ten, but what if not all events were actually recorded? This raises a concern with the tool's accuracy and completeness. Another example, are the timestamps that were recovered accurate to the recorded activity's timestamps? An evaluation of the two apps' accuracy and completeness can be produced here.
3. **The correlation of forensics in cross-applications**: To establish connections between recovered artifacts from all apps used by the group and be able to combine them into a shared timeline of events. ALEAPP's recovered findings should be validated against the documented test activities, examining the tool's limitations.

2.2 Hoang's Selection of Tool, Apps, and Novelty

Tool: Avilla Forensics. This tool is free to use and works through ADB and can extract data of apps from non-rooted devices using an APK downgrade technique installing an older version of the app, making it possible to capture the data through Android backup.

Apps:

- Telegram: A cloud-based messaging application to send messages, share files, and for communication.
- Firefox: An open-source web browser to search the internet and download files.

Novelty:

1. **Non-rooted Forensic Acquisition:** To explore the tool's APK downgrade technique, obtaining the two apps' data without requiring root access.
2. **Forensic Analysis of Novel Applications**: To investigate Telegram's communication artifacts and the browsing history of Firefox, including timestamps and other available data.

3. **Evaluation of APK Downgrade Limitations**: Examining how the downgrade technique affects the data of applications, what information can be recovered, and if data is lost or modified during the process.

3. Integrated Work Component

We plan to combine our findings into a shared timeline of events to reconstruct the activities of the suspect from all four Android applications that we will be using. This involves notes from Joplin, GPS activities from OpenTracks, messages from Telegram, and browsing history (including timestamps) from Firefox. We will be comparing how we acquired data since Rafaell will be using a rooted Android emulator with ADB and ALEAPP, while Hoand will be using Avilla Forensics on a non-rooted emulator. We will be aiming to identify the differences in these said processes, the evidence that have been recovered, and its limitations.

4. Use of AI

To assist in the development of the project, Rafaell will be primarily using ChatGPT while Hoang will be using Claude.

Tool	Version and Account Type	Purpose
ChatGPT	GPT-6, Plus	To plan this project, to understand concepts, guidance in technical aspects, to troubleshoot, assist in documents
Claude	Opus 5.5, Pro	Same purpose as ChatGPT, included to have another perspective/AI standpoint

Table 4.1 List of AI Tools Used, Version, and Purpose

4.1 Value Addition

We discussed later on that when we prompt with our respective AI tool, we should inquire for main ideas on whatever it is that we are needing assistance with. From getting these main ideas, we are correctly guided on what we need to include and what we need to do without much need of elaboration. In essence, we ask for guidance instead of the complete answer, allowing the both of us to still do our own research elsewhere and make our own decisions. By doing it this way, the use of AI tools serve as a support to us, instead

of letting it do the work for us, because we believe that learning should still come from our own effort.

Appendix

Appendix A: Work Log Table of Rafaell

Date	Student	Number of Hours	Description of work
10-9-2026	Rafaell	1	Read the course project intro document, brainstormed ideas
10-15-2026	Rafaell	1	Continued brainstorming ideas
10-16-2026	Rafaell	2.5	Initialized the project, started the project proposal and met with the professor to introduce the project
10-17-2026	Rafaell	3	Examined and tested ALEAPP in extracting data from apps including Joplin, and OpenTracks
10-24-2026	Rafaell	3	Continued working on the Project Proposal
10-26-2026	Rafaell	1	Finalized Project Proposal for submission

Appendix B: Work Log Table of Hoang

Date	Student	Number of Hours	Description of work

10-9-2026	Hoang	1	Read the course project document, brainstormed ideas
10-15-2026	Hoang	1	Continued brainstorming ideas
10-16-2026	Hoang	2.5	Initialized the project, started the project proposal and met with the professor to introduce the project
10-17-2026	Hoang	3	Researched Avilla Forensics including supported apps, APK downgrade
10-24-2026	Hoang	3	Continued working on the Project Proposal and shared progress
10-25-2026	Hoang	1	Finalized Project Proposal

Appendix C: Project Contract

The members of Group 6 agree on the following statements to ensure progress is made together for the completion of the project on time.

1. To be regularly in touch and updated with each other about any progress made via communication apps or in-person.
2. To meet at least once a week (Fridays, right after the course class) to continue discussion of progress.
3. To be individually responsible for researching, testing, and documenting the chosen tools and applications.
4. To work and help each other in completing the project, creating the shared timeline of events, all of the project reports, the video demonstration, and the final presentation.
5. To keep track of the work hours done for the project and update the work log table.
6. To document the use of AI and save important prompt histories.
7. To upload individual work to the GitHub repository, submit documents before the deadline, and finish assigned tasks on time.

By signing with our name initials below, we agree to follow this contract.

Team Member	Signature	Date
Rafaell Carlos Gramata	RCG	10-26-2026
Hoang Nghiem	HN	10-26-2026

Appendix D: References

Android Developers. (n.d.). *Extended controls, settings, and help*. https://developer.android.com/studio/run/emulator-extended-controls

Brignoni, A., & contributors. (n.d.). *ALEAPP: Android Logs Events and Protobuf Parser* [Computer software]. GitHub. https://github.com/abrignoni/ALEAPP

Joplin. (n.d.). *Joplin*. https://joplinapp.org/

Mozilla. (n.d.). *Firefox*. https://www.mozilla.org/firefox/

OpenTracks contributors. (n.d.). *OpenTracks* [Mobile application]. GitHub. https://github.com/OpenTracksApp/OpenTracks

Telegram. (n.d.). *Telegram Messenger*. https://telegram.org/

Appendix E: Prompt History

Our conversations with AI Tools.

ChatGPT

Prompt

Analyze each page of these documents about a project carefully. I want to use ALEAPP as my tool and my teammate may be using Zalo and Discord.

Response

I've carefully reviewed both pages of your uploaded CSIS 4440 Project Proposal Template.

This template is especially important because it contains additional requirements that weren't clearly specified in the earlier project guidelines, including a signed team contract, detailed individual work logs, and an explanation of the originality and added value of your investigation.

Your group is pursuing Option 1: Existing Forensic Tools and Novel Android Application Analysis, so I'll explain each requirement in relation to your actual project.

1. Page 1 — Proposal structure and required content

A. Submission requirements

The document states:

Requirement	Details
Submission type	One joint proposal per group
Your group	Group 6
Filename	F26_4440_G6_Proposal.pdf
Submission platforms	Blackboard and GitHub
Submitted by	Designated team lead
Length	At least 2 pages
Writing quality	Grammar, structure, and professional presentation are graded

Important discrepancy: Your previous project guidelines specified a 1–2-page proposal, but this template explicitly says at least 2 pages.

I recommend aiming for two complete pages of main proposal content, with supporting sections and appendices as needed. The template does not specify whether appendices count toward the minimum.

B. Title and group identification

The title section must contain both students' full names, student IDs, and a clearly designated team lead.

For your group:

Member	Student ID
Rafaell Carlos Gramata	300405898
Hoang Nghiem	300416011

You'll need to identify which of you is the team lead.

A possible working title is:

Android Forensic Reconstruction of Communication, Planning, and Location Artifacts

This reflects your shared investigation without implying that you can already prove a suspected transaction occurred.

C. Tools and Apps Overview — Each student

This is one of the most important parts of the template.

The professor explicitly requires you to identify what each student will individually explore, understand, and apply.

	Your investigation	Teammate's investigation
Forensic tool	ALEAPP	Avilla Forensics
Application 1	Joplin	Zalo
Application 2	OpenTracks	Discord
Evidence	Notes, attachments, timestamps, GPS tracks	Communication artifacts, messages, attachments, timestamps
Progress	Joplin extraction and analysis successful	Feasibility not yet confirmed

For your part, you've already demonstrated that ALEAPP can process Joplin's SQLite database and recover notes and notebook information.

For OpenTracks, the relevant ALEAPP modules exist, but an actual route extraction and analysis still needs to be demonstrated.

Your teammate's forensic acquisition and interpretation capabilities must also be verified.

D. Novelty and value addition

The template asks two related questions:

What makes your proposed project novel?

What value will you add beyond what the forensic tools already provide?

Simply explaining that ALEAPP is not on the class list is a starting point, but not a complete answer to the second question.

For your group, possible value additions include developing a controlled investigation scenario, validating extracted records against known actions, normalizing timestamps, and comparing evidence from multiple applications.

Your planned integrated timeline could be a particularly strong addition.

E. Integrated component

The professor explicitly encourages integration of both students' contributions.

The template gives examples such as a jointly generated automated report or an integrated tool that combines the students' analyses.

Your proposed cross-application timeline fits the broader integration objective, but the template's examples emphasize technical integration, not merely placing two separate investigations in the same report.

You could explain that the group intends to combine verified findings from Joplin, OpenTracks, Zalo, and Discord into one chronological analysis, while leaving automation as a possible enhancement.

F. AI Use Section

The required table must document:

AI Tool Name	Version and Account Type	Specific Use
ChatGPT	GPT-6, Plus account	Project planning, explaining forensic concepts, technical troubleshooting, and writing assistance
Other AI tools, if used	Actual version and account type	Specific tasks performed

Only include tools actually used, and distinguish their contributions accurately.

The table alone isn't enough. Page 2 also requires your personal value addition and prompt history.

2. Page 2 — Individual accountability and documentation

This page contains several requirements that could easily be overlooked.

A. Value Addition beyond AI

The template emphasizes that each student must personally explain the value they contributed beyond AI assistance.

For your ALEAPP investigation, this could include work you actually performed:

Installing and configuring ALEAPP.

Setting up an Android emulator with root access.

Installing Joplin and OpenTracks.

Locating Joplin's SQLite databases.

Acquiring application artifacts using ADB.

Calculating the SHA-256 hash of your evidence.

Running ALEAPP and examining its recovered notes and timestamps.

These are real activities from your investigation that you can document.

You should eventually explain how you interpreted and validated the findings yourself, rather than simply saying you followed AI-generated instructions.

B. AI prompt history

The template explicitly states that all AI prompt history must be saved in the appendix.

That means your ChatGPT conversations used for project planning, setup, troubleshooting, analysis, and writing need to be preserved.

For the proposal, I recommend organizing the relevant history chronologically and identifying which conversations informed the work described in the proposal. The template doesn't specify an approved format for exporting prompts.

C. Signed project contract

This is a new, explicit requirement.

Both students must sign an agreement describing how the team will work together.

The professor suggests including meeting frequency, communication methods, and any other agreements relevant to the project.

For your team, the contract could cover the following:

Area	Suggested agreement
Meeting frequency	Meet at least once weekly
Communication	Microsoft Teams or another agreed channel
Individual responsibilities	Each student owns their forensic tool and two applications
Documentation	Both maintain their own work logs
Integration	Both contribute verified findings to the shared timeline
Deliverables	Team lead coordinates final submissions
Problems or delays	Inform the other member promptly
Signatures	Both students sign and date the contract

These are suggested terms, not requirements beyond the signed agreement requested in the template.

D. Individual work logs

The professor places particular emphasis on accurate, frequent documentation of work.

The table shown on Page 2 contains four columns:

Date	Student Name	Number of Hours	Description of Work
Actual date	Student who performed the work	Actual time spent	Specific activities completed

The template explicitly warns against broad entries showing 10 or 12 hours of work in one day without sufficient detail.

For your own contribution, you should reconstruct your earlier ALEAPP setup sessions using available command history, screenshots, and other evidence. Record only hours you can reasonably substantiate.

Your previous project introduction also establishes an expectation of at least 60 hours per student throughout the term. That doesn't mean 60 hours must be completed by the proposal deadline.

E. Closing and references

The proposal must list literature and sources cited in the document.

For your investigation, appropriate sources could include official ALEAPP documentation, the application repositories, Android developer documentation, and technical references on relevant forensic artifacts.

The template doesn't specify a mandatory citation style.

3. What the proposal should demonstrate about your project

I would organize your proposal around these questions:

Proposal component	What the professor should understand
Project title	What investigation you are conducting
Forensic motivation	Why the evidence matters in the simulated incident
Student 1	What you will investigate using ALEAPP
Student 2	What your teammate will investigate using Avilla
Novelty	How your chosen tools and apps extend beyond class topics
Value addition	What investigation and analysis you contribute beyond existing tool functionality
Integration	How the two investigations will produce a shared chronology
Feasibility	Which capabilities are already demonstrated and which still need testing
AI use	How AI assisted and what work you performed independently
Team contract	How the group agrees to collaborate
Work logs	What each member has actually done so far
References	Sources supporting your technical plan

4. My assessment of your group's current readiness

Proposal preparation checklist

Chosen project option	Option 1
Tools and applications selected	Tentative selections identified
Forensic scenario	Simulated suspicious transaction
Joplin forensic feasibility	Successfully tested
OpenTracks forensic feasibility	Still needs testing
Zalo and Discord feasibility	Not yet demonstrated
Integrated timeline	Proposed, not implemented
Team contract and work logs	Need to prepare/confirm

5. What you need to prioritize before October 25

Your proposal and GitHub repository are due Sunday, October 25, 2026, at 11:59 PM.

I'd prioritize the following:

Confirm the final tools, applications, project scenario, and team lead.

Write separate, clearly defined sections for each student's work.

Explain the investigation's novelty and added value.

Describe how you intend to correlate findings into a shared timeline.

Prepare and sign the project contract.

Complete individual work logs and AI-use documentation.

Include the AI prompt-history appendix and references.

Submit the PDF on Blackboard and commit it to the required GitHub repository.

Overall assessment

Your project direction is compatible with the proposal template, and your successful Joplin examination gives you a credible starting point.

However, the proposal must do more than list ALEAPP, Avilla, and four Android applications. It needs to communicate each student's independent investigative contribution, why the examination is valuable, and what additional analysis the group will perform beyond simply running the tools.

The biggest items you shouldn't overlook are the signed project contract, individual daily work logs, AI-use explanation, and full prompt-history appendix. These are explicitly required by the template and can be prepared even while some technical experiments are still underway.

Prompt

based on the documents, in starting the project intro document, what are the things i should draft and briefly describe what to put in them. only give me ideas

Response

Based on your CSIS 4440 Project Introduction and Proposal Template, here are the main sections you should prepare for your project proposal.

1. Project Title and Group Information: Include your project title, group number, team members' names and student IDs, and identify the team lead.

2. Project Overview: Briefly explain what your project is about, the forensic scenario, and what you want to investigate.

3. Tools and Applications: Identify the forensic tool and two Android applications assigned to each student. Briefly describe their purposes and what data you plan to examine.

4. Novelty and Value Addition: Explain what makes your chosen tools and applications new to the class and what additional investigation or analysis you plan to contribute.

5. Integrated Work Component: Describe how both members plan to combine their findings, such as creating a shared timeline of events.

6. AI Use: List the AI tools used, their versions, account types, and specific purposes. Explain what work you personally contributed beyond AI assistance.

7. Project Contract: Include agreements about communication, meetings, responsibilities, deadlines, and teamwork. Both members must sign it.

8. Work Logs: Record each member's actual work dates, hours spent, and tasks completed.

9. References and Appendix: Include sources used for research and preserve the AI prompt history.

These sections follow the proposal template. You can begin by writing short descriptions under each heading and expand them once you and your teammate finalize the details.

Prompt

How to i install aleapp, joplin, and opentracks

Response

Here's how to install ALEAPP, Joplin, and OpenTracks for your CSIS 4440 forensic investigation, using the same setup we successfully used before.

Important: ALEAPP runs on your Windows computer, while Joplin and OpenTracks are installed inside your Android emulator.

1. Install ALEAPP on Windows

1. Visit the official ALEAPP GitHub Releases page.

2. Find the release you want and expand Assets.

3. Download the Windows 64-bit installer, ending in windows-x64-setup.exe.

4. Verify that it came from the official release and check its SHA-256 checksum if provided.

5. Run the installer and follow the setup instructions.

6. Launch ALEAPP.

Version you previously used: ALEAPP 2026.4.3.

After installation, the ALEAPP interface should show options for selecting an evidence source, output folder, and forensic analysis modules.

2. Prepare the Android emulator

In Android Studio:

1. Open Device Manager.

2. Create a Pixel 6 virtual device.

3. Select an Android Google APIs x86_64 system image, rather than a Google Play image.

4. Launch the emulator.

We originally used an Android 14 (API 34) Google APIs image for the walkthrough.

In Windows Command Prompt, verify ADB access:

adb devices

adb root

adb shell id

Successful root access will display:

uid=0(root) gid=0(root)

This allows you to acquire protected application files later.

3. Install Joplin

Joplin

Version used in your project: 3.7.10

Download an Android APK from the application's official distribution sources, rather than the Windows desktop installer.

You previously installed:

joplin-v3.7.10.apk

Open Command Prompt in the folder containing the APK, then run:

adb install joplin-v3.7.10.apk

Expected result:

Performing Streamed Install

Success

Verify the package:

adb shell pm list packages | findstr joplin

Expected:

package:net.cozic.joplin

Open Joplin on the emulator to ensure it launches normally.

4. Install OpenTracks

OpenTracks

Version used in your project: 4.22.0

Obtain a compatible APK from a trusted OpenTracks distribution source.

You previously used:

de.dennisguse.opentracks.playstore_v4.22.0.apk

With Command Prompt open in the APK's folder, run:

adb install de.dennisguse.opentracks.playstore_v4.22.0.apk

Expected output:

Performing Streamed Install

Success

Verify the installation:

adb shell pm list packages | findstr opentracks

Expected:

package:de.dennisguse.opentracks.playstore

Open OpenTracks in your emulator and confirm that it launches correctly.

5. Verify both applications

Run:

adb shell pm list packages | findstr /i "joplin opentracks"

For your setup, the output was:

package:net.cozic.joplin

package:de.dennisguse.opentracks.playstore

Summary of your setup

Software	Installed where?	Version
ALEAPP	Windows laptop	2026.4.3
Joplin	Android emulator	3.7.10
OpenTracks	Android emulator	4.22.0
ADB	Windows, through Android SDK Platform Tools	SDK-dependent

For your project report: Take screenshots of ALEAPP, the two installed Android apps, the successful adb install commands, and the package verification output. These will help document your setup and demonstrate that you configured the forensic environment yourself.

Once installed, the next phase is generating controlled application activity, acquiring its evidence, and processing it with ALEAPP.

Prompt

i want to simply test if this tool can extract data from these two apps, guide me in the extraction process. i will be creating a note in joplin, and how should the location activities for opentracks work if i will only be working on an emulator? keept is short, only a summary and not step-by-step

Response

To test ALEAPP with both applications, you can follow these steps:

1. Joplin: Create a sample note and save it. Use ADB to extract Joplin's application data from the emulator, then analyze it using ALEAPP's Joplin modules.

2. OpenTracks: Since you're using an Android emulator, you can simulate movement through Android Emulator's Extended Controls → Location → Routes. Create a route, start recording in OpenTracks, and play the simulated route.

3. Data Extraction: After generating data in both apps, use ADB to acquire their application files. Keep copies of the original extracted data.

4. ALEAPP Analysis: Load the extracted files into ALEAPP and select the corresponding modules for Joplin and OpenTracks.

5. Verification: Check whether ALEAPP recovered your sample note, GPS tracks, coordinates, and timestamps. Compare the findings with the activities you performed.

This should help determine whether ALEAPP can successfully analyze data from both applications and identify any limitations during the process.

Prompt

how do i make a project contract, how do i start it and give me some ideas that my group should keep in mind of and agree. only give me ideas, keept it short and dont draft them for me

Response

For your CSIS 4440 project contract, start with a short statement explaining that both group members agree to work together and complete their responsibilities throughout the project.

Here are some things you and your teammate should discuss and agree on:

1. Communication: How will you contact each other? For example, Discord, Teams, or WhatsApp.

2. Meetings: How often will you meet to discuss your progress and problems?

3. Individual responsibilities: Each member should be responsible for researching, testing, and documenting their assigned forensic tool and applications.

4. Shared work: Decide how you'll divide the integrated timeline, reports, and final presentation.

5. Work logs: Agree to regularly record your work dates, hours, and completed tasks.

6. Deadlines: Decide when each member should finish their tasks so there's enough time to review everything.

7. Problems: Agree on what to do if someone falls behind or experiences technical difficulties.

You can end the contract with a statement such as:

By signing below, we agree to follow these terms and contribute fairly to our group project.

Include both members' names, signatures, and dates.

Make sure you and your teammate discuss and agree on the terms before signing the contract.

3.3.2.2 Claude

Prompt

“In a fictional forensic activity about a sale of illegal goods, what activities should I make in telegram and firefox to examine later?"

Response

1. Harmless activity to create in Telegram and Firefox

Keep everything fictional. Use made-up names and a vague, harmless story, like arranging to hand over a “package” at a public place. Write down the time of each action.

Telegram (two test accounts, “Seller” and “Buyer”)

Seller sends: “Got the package. Can we meet tomorrow?” Buyer replies: “Yes, 3pm at the library parking lot.” Seller sends a photo (any harmless image). Seller sends a message, then deletes it. Buyer sends a final message: “Bring cash. Delete this chat after.” Optionally, edit one message once.

Firefox

Search for “meeting place near [your city]” and open two of the results. Visit a map page and search the library address. Bookmark one page. Download one harmless file (a PDF or image). Open one page in a private tab, so you can see whether it leaves any trace.

Ground-truth log: Time, app, action, exact text or URL. You’ll compare Avilla’s results against this later.

Prompt

What should I record before and after using Avilla so i can reuse?

Response

Before

- Date, time, and time zone
- Avilla Forensics version
- Emulator name, Android version, and system image
- Telegram and Firefox versions (`adb shell dumpsys package <package> | findstr versionName`)
- The settings you enabled and the exact steps you plan to follow
- A screenshot of the emulator state and the final ground-truth log

After

- Every step and option you chose in Avilla
- Any errors or warnings, copied word for word

- Output file names, sizes, and locations
- SHA-256 hashes of the output files (a `hasher` tool or PowerShell's `Get-FileHash`)
- The app versions after the run, to show whether downgrade and restore changed anything
- What was recovered versus what you expected from your ground-truth log
- Screenshots of the key steps

Prompt

What does avilla forensics need before I do an acquisition?

Response

Check the tool's own page and README for exact requirements, since this is a general guide.

- **USB debugging on.** In the emulator, turn on Developer options, then USB debugging. Confirm the emulator shows up when you run `adb devices`.
- **Unlocked device.** Keep the screen on and the emulator unlocked.
- **Android version.** Android 12 and newer stop including app data in standard backups for most apps, so an older version like Android 11 is more likely to work with the downgrade method. Test it before relying on it.
- **Apps installed and supported.** Avilla's page lists Telegram and Firefox among its APK downgrade apps. Note the exact app versions you have.
- **Backup prompt.** adb backup usually shows a confirmation screen on the device. Be ready to tap it, and leave the password blank unless Avilla says otherwise.
- **Enough storage.** Make sure the emulator has free space for the downgraded apps and the backup file.

A working ADB. Install the platform tools on Windows and check the tool's connection works before the real run.